

Name : _____

Student Number: _____

Section: _____

Duration: 50 minutes**Instructions:**

1. A calculator, a self-prepared formula sheet (up to half an A4 page) and a package of Statistics tables are allowed. **No other aids** such as books, or notes are permitted.
 2. Questions 1 through 9 are short-answer questions. You must show some work on your question paper for full credits. **ANSWER ONLY** will receive a score of Zero.
 3. Each short-answer question is worth 3 marks. If your final answer is incorrect **only** due to a **minor** error, 2 marks will be awarded. If there is a significant error, but some progress towards the solution, 1 mark will be awarded.
 4. Question 10 is a full-answer question. Write out your solution carefully and completely on the examination paper. Part-marks will be awarded for partial work for this question. **Marks will be deducted for incomplete or poorly presented solutions.**
 5. For numerical answers, please give your answer to at least 4 decimal places.
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Question	1-9	10	Total
Mark Received			
Out of:	27 marks	6 marks	33 marks

(Questions 1 and 2 refer to the following scenario): In a survey of 500 electrical engineering students at a university, it is discovered that 320 of them own a personal oscilloscope. Let p represent the proportion of electrical engineering students at the university who own a personal oscilloscope.

1. Compute the p-value for the hypothesis test $H_0 : p = 0.6$ vs $H_1 : p > 0.6$.

$$\hat{p} = \frac{320}{500}$$

$$Z_{obs} = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}} = \frac{0.64 - 0.6}{\sqrt{0.6(1-0.6)/500}} = 1.83$$

$$p\text{-value} = P(Z > 1.83)$$

$$= 1 - P(Z \leq 1.83)$$

$$= 1 - 0.9664$$

$$= 0.0336$$

0.0336

2. **Do not** use the above data as a pilot study. Determine the number of students needed to estimate p within 0.03 with 94% confidence.

$$n = \frac{Z_{\alpha/2}^2}{4E^2}$$

$$= \frac{1.88^2}{4 \cdot 0.03^2}$$

$$= 981.78$$

$$n = 982$$

$$E = 0.03 \quad Z_{\alpha/2} = Z_{0.03}$$

$$= 1.88$$

$\hat{p}$ is unknown.

982

Questions 3 and 4 refer to the following scenario: A manufacturer claims its light bulbs last 1200 hours on average. A sample of 25 bulbs averages 1180 hours with a 50-hour standard deviation. Is there evidence that the true mean lifespan μ differs from the claimed 1200 hours? (Assume normal distribution)

3. **State the hypotheses:** H_0 VS H_1 and compute the **observed value** of the test statistic if H_0 is true.

$$H_0: \mu = 1200 \quad H_1: \mu \neq 1200$$

$$t_{obs} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{1180 - 1200}{50/\sqrt{25}} = -2$$

-2

4. Compute the p-value.

Since $H_1: \mu \neq 1200$

$$P\text{-value} = 2 \cdot P(t_{24} > |z_{obs}|)$$

$$= 2 \cdot P(t_{24} > 2)$$

$$0.025 < P(t_{24} > 2) < 0.05$$

$$0.05 < P\text{-value} < 0.1$$

$0.05 < P\text{-value} < 0.1$

5. A manufacturer compares two soldering techniques to see if the new technique reduces defective boards. In a sample of 800 boards assembled using the current technique, 56 were found to be defective. In a sample of 900 boards assembled using the new technique, 42 were found to be defective. Conduct a hypothesis test to determine if the new technique results in a lower proportion of defective boards.
- $\hat{p}_1 = \frac{56}{800} = 0.07$
 $\hat{p}_2 = \frac{42}{900} = 0.0467$
- $H_0: p_1 - p_2 \leq 0$ $H_1: p_1 - p_2 > 0$
- Compute the observed value for the hypothesis test above.

$$Z = \frac{(0.07 - 0.0467) - 0}{\sqrt{\frac{0.07 \cdot 0.93}{800} + \frac{0.0467(1-0.0467)}{900}}} = 0.0233 / 0.01144 = 2.0367$$

2.04

6. An engineer investigates the effect of a new cooling system on the temperature of processors. The temperatures (in °C) of six processors are measured before and after installing the new system. The following table presents the observations:

Processor	1	2	3	4	5	6
Before	85	90	88	92	87	89
After	78	84	80	85	82	83
Differences	7	6	8	7	5	6

Each difference represents the temperature before minus the temperature after installing the new cooling system. Let μ_D represent the population mean of all such differences.

Compute the 95% confidence interval for μ_D .

$\bar{d} = 6.5$ $Sd = 1.0488$ $t_{0.025, 5} = 2.571$

$$\bar{d} \pm t_{\alpha/2, n-1} \frac{Sd}{\sqrt{n}}$$

$$= 6.5 \pm 2.571 \frac{1.0488}{\sqrt{6}}$$

$$= 6.5 \pm 1.1008 = (5.3992, 7.6008)$$

Questions 7 to 9 refer to the following setup: A mechanical engineering team conducted a study to examine the relationship between the diameter of a pipe (X) and the flow rate of a fluid through the pipe (Y), measured in liters per minute. Data was collected from **15** different pipes, and the following statistics were calculated:

$$\sum x_i = 300 \quad \sum y_i = 4,500 \quad \sum x_i y_i = 106,300 \quad s_x = 10.5 \quad s_y = 120.3$$

Assume both X and Y follow normal distributions

7. Calculate the coefficient of determination.

$$\begin{aligned} S_{xy} &= \frac{1}{n-1} \left[\sum xy - \frac{\sum x \cdot \sum y}{n} \right] \\ &= \frac{1}{14} \left(106,300 - \frac{300 \cdot 4,500}{15} \right) \\ &= 1164.2857 \end{aligned}$$

Answer:

$$0.8496$$

$$r = \frac{S_{xy}}{s_x \cdot s_y} = 1164.2857 / (10.5 \cdot 120.3) = 0.9217 \quad r^2 = 0.9217^2 = 0.8496$$

8. Determine the equation of the regression line.

$$b_1 = \frac{1164.2857}{10.5^2} = 10.56$$

Answer:

$$\begin{aligned} b_0 &= \bar{y} - b_1 \bar{x} = 300 - 10.5 \cdot 20 \\ &= 88.8 \end{aligned}$$

$$\hat{y} = 88.8 + 10.56 \cdot x$$

$$\hat{y} = 88.8 + 10.56x$$

9. Test the hypotheses $H_0 : \rho = 0$, $H_a : \rho \neq 0$ for a linear relationship between X and Y . Compute the p-value. *and the disty of the test stats if H_0 is true.*

$$t = r \cdot \sqrt{\frac{n-2}{1-r^2}} \sim t_{n-2}$$

Answer:

$$t_{obs} = 0.9217 \cdot \sqrt{\frac{15-2}{1-0.9217^2}}$$

$$= 8.567 \sim t_{13}$$

This is a full-answer question.

Write out ALL of your work to get full marks.

An experiment compared the battery life of two types of batteries: Lithium-ion and NiMH. 20 samples of Lithium-ion and 15 samples of NiMH were tested. The results are as follows:

Lithium-ion	$\bar{x}_1 = 450$ hrs	$s_1 = 21$ hrs	$n_1 = 20$
NiMH	$\bar{x}_2 = 400$ hrs	$s_2 = 25$ hrs	$n_2 = 15$

Is there evidence that the Lithium-ion battery life exceeds that of the NiMH battery? Assume normal distribution. Let $\mu_1 - \mu_2$ represent the difference in population means for Lithium-ion and NiMH.

(a) (1 mark) State the hypotheses.

$$H_0: \mu_1 - \mu_2 \leq 0 \quad H_1: \mu_1 - \mu_2 > 0$$

(b) (1 mark) State the test statistic you will use. What **distribution** (including degrees of freedom, if appropriate) you will use to calculate the p-value.

$$s_2/s_1 = 25/21 = 1.19 \leq 1.4 \Rightarrow \text{pooled } t$$

$$T = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)_0}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \sim t_{33}$$

(c) (2 marks) Find the observed value of the test statistic.

$$t_{obs} = \frac{(450 - 400) - 0}{\sqrt{519.0606 \left(\frac{1}{20} + \frac{1}{15} \right)}} = \frac{50}{7.7818} = 6.4252$$

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = 519.0606$$

(d) (1 mark) Compute (or bracket) the p-value within the accuracy of the tables.

$$p\text{-value} = P(t_{33} > 6.4252)$$

$$p\text{-value} < 0.005$$

(e) (1 mark) If we were testing our hypothesis at the level $\alpha = 0.05$, would we reject H_0 ? Explain why or why not. In order to receive a mark, you must provide a (correct) explanation

$$\text{since } p\text{-value} < 0.005 < \alpha = 0.05$$

We reject H_0